SOFTWARE DEVELOPMENT PORTFOLIO

MATHAVAN P

Software Developer | Full-Stack Project Experience

👉 **Click to View Live Work: https://mad-eats-eight.vercel.app**

PROFESSIONAL PROFILE

Computer Science student with hands-on experience building, containerizing, and deploying full-stack web applications. Interested in building reliable cloud applications, developing RESTful APIs, managing database workflows, and solving practical software engineering problems. Continuously developing skills across modern web development, backend architectures, and automated cloud pipelines.

TECHNICAL SKILLS

Programming	Java (21 LTS), JavaScript (ES6+), SQL, HTML5, CSS3
Backend	Spring Boot 3, Spring Data JPA, Hibernate, REST API Development
Frontend	React.js, Vite, Tailwind CSS, State Management, Responsive Design
Database	MySQL, TiDB Cloud Serverless MySQL, HikariCP Connection Pooling
Cloud & DevOps	Docker Containerization, Vercel Edge CDN, Render Cloud, Multi-Stage Builds
Messaging & Docs	Resend API (HTTPS Port 443), Automated PDF Invoice Generation (OpenPDF)
Collaboration	Git, GitHub, Maven (mvnw), Agile Version Control Workflows

PROJECT EXPERIENCE

MAD EATS — Online Food Ordering Platform

FULL-STACK DEVELOPER

Architected and deployed a complete, live cloud-based food ordering system with customer storefront, dynamic cart, automated invoice compilation, and instant kitchen email alerts.

- Developed full-stack web application with a **React** frontend and **Java 21 Spring Boot** backend.
- Implemented categorized menu browsing (16 dishes across 8 categories), shopping cart, and Cash on Delivery (COD) checkout.
- Engineered an automated **dynamic PDF invoice & receipt generator** producing itemized bills on order placement.
- Integrated the **Resend API** over HTTPS (port 443) to deliver real-time order alerts with attached PDF invoices to the owner.
- Migrated and managed production database workflows on **TiDB Cloud Serverless MySQL** with SSL encryption.
- Containerized backend using **Docker** and deployed to **Render** and frontend to **Vercel** for 24/7 zero-cost uptime.
- Configured Vercel reverse-proxy rewrites to eliminate CORS errors and achieve seamless same-origin API latency.

Java 21 | Spring Boot | React | MySQL / TiDB Cloud | Docker | Resend API | Vercel | Render | Git / GitHub

PROJECT OVERVIEW & LIVE DEPLOYMENT

LIVE CUSTOMER WEBSITE
https://mad-eats-eight.vercel.app

LIVE CLOUD BACKEND API
https://mad-eats-backend.onrender.com

GITHUB SOURCE CODE
https://github.com/Mathavan-27/mad_eats

CLOUD INFRASTRUCTURE MODEL
100% Free Tier • 24/7 Uptime • Zero Server Cost

SYSTEM ARCHITECTURE FLOW

Customer UI (React on Vercel) → **REST API** (Spring Boot on Render) → **Cloud MySQL** (TiDB Serverless) → **PDF Engine** (OpenPDF Receipt) → **Owner Alert** (Resend HTTPS API)

KEY CAPABILITIES DELIVERED

Direct Customer Ordering
Allows restaurants to accept orders directly online without paying 30% commissions to food aggregators, with instant cart updates and Cash on Delivery support.

Instant Kitchen Dispatch
Automated HTTPS alert pipeline dispatches real-time emails containing customer phone, address, and items with attached PDF invoices within seconds of checkout.

Dynamic Digital Invoicing
Compiles printable, branded invoice receipts on the fly featuring unique order codes, tax breakdowns (5% GST), and delivery fees for customers and kitchen staff.

24/7 Zero-Maintenance Cloud
Operates autonomously on distributed cloud tiers (Vercel CDN, Render Docker containers, TiDB Cloud database) with continuous availability and zero local machine dependency.

ADDITIONAL TECHNICAL WORK

Alongside this live commercial project, I continuously build independent software projects to strengthen my engineering skills in full-stack architectures, microservices, database performance, and cloud infrastructure. Additional codebases and repositories are accessible on my GitHub profile.

DEVELOPER
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